

Mid Semester Examination 2025

Name of the Program: B.Tech Petroleum Engineering

Semester: VI

Name of the Course: Natural Gas Engineering

Course Code: TPE-603

Time: 1-1/2 Hour Maximum

Marks: 50

Note:

(iii) Answer **all the questions** by choosing *any one of the sub questions*.

(iv) Each question carries 10 marks

Q1	(10 marks)		CO 1, CO 2											
(a)	Explain in detail about the conventional natural gas production system? How do you differentiate the gas well, condensate well and oil well? Discuss associated gas, non-associated gas and condensate gas? How they are different from each other													
OR														
(b)	Explain the phase diagram on PV curve? What do you mean by quality lines and retrograde condensation? Explain with the help of PT diagram?													
Q2	(10 marks)		CO 2											
(a)	What do you mean by fracturing fluid? What are the functions of fracturing fluids? How fracturing fluid is made, Discuss all the materials being used in making them.													
OR														
(b)	Calculate the gas specific gravity of natural gas in the reservoir having the following composition.													
<table><tr><th>Composition</th><th>Mole fractions</th></tr><tr><td>C1</td><td>0.775</td></tr><tr><td>C2</td><td>0.083</td></tr><tr><td>C3</td><td>0.021</td></tr><tr><td>CO2</td><td>0.030</td></tr><tr><td>H2S</td><td>0.020</td></tr></table>		Composition		Mole fractions	C1	0.775	C2	0.083	C3	0.021	CO2	0.030	H2S	0.020
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The reservoir is having at initial condition of $P_i = 3500\text{psia}$ and 185°F . By calculating Sp. Gr. Of the gas also calculate T_{pc} , P_{pc} , T_{pr} , P_{pr} by using Standing and Katz correlations (take compressibility factor $Z=0.87$ for the data given) and hence the density of natural gas														
Q3	(10 marks)		CO 1											
(a)	By which phenomena CBM is trapped in the bed seam? Compare Coal bed methane (CBM) and Conventional reservoir on the different aspects? Discuss the important needs of increased CBM production													
OR														

(b)	What is natural gas? Discuss all the properties with mathematical expressions of Natural gas.	
Q4	(10 marks)	
(a)	Define proppant? What are the functions of proppant and discuss the types of proppant? Discuss proppant selection considerations	CO 1, CO 2
OR		
(b)	With the help of diagram explain the extraction of natural gas by the enhanced (CBM) techniques from CBM? Write down the usage and advantages of CBM.	
Q5	(10 marks)	
(a)	Discuss Dehydration of Natural gas in details? What are the effects of presence of water in natural gas?	CO 1, CO 2
OR		
(b)	How CBM well different from the natural gas well, explain with the help of net production vs. time graph? Discuss in detail about the significance of CBM in Indian Energy basket?	